

Bachelor Thesis Project

Network Analysis of Global Financial Markets

Bachelors' Degree in
Mathematics and Computing

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Introduction

- Financial markets are inherently complex systems in which thousands of individual assets interact.
- A network based approach helps in analysing large amount of time-series stock market data by making it easier to visualise.

Vision:

- To model and interpret financial market interdependencies using network theory for better risk insight and diversification.

Mission Statement:

- To visualize, and quantify relationships across Indian, U.S., and global financial markets.
- To track the evolution of market structure across time and crises
- To identify the most influential stocks

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Ideation

Topological Methods

Those properties of a geometric object which remain unchanged on deformation.

Network Properties

- Average Correlation
- Network Density
- Clustering Coefficient
- Characteristic Path length

Graph Centralities

- Betweenness Centrality
- Closeness Centrality
- Degree Centrality
- Eigen Vector Centrality
- PageRank

Minimum Spanning Tree

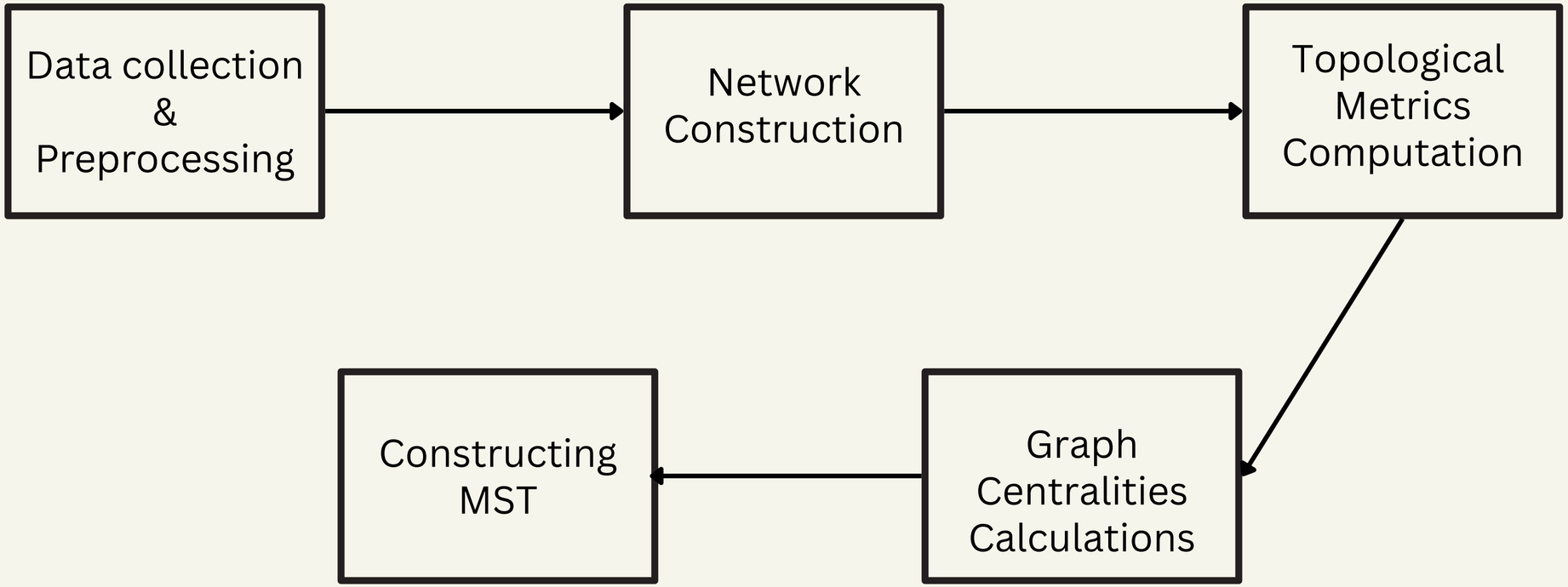
A MST is a subset of the edges of a connected, edge-weighted undirected graph that connects all the vertices together, without any cycles and with the minimum possible total edge weight.

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- The first phase focused on analysing Indian and U.S. stocks over 2006-2025
- The second phase focused global market analysis over 2018-2025





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Data Collection & Preprocessing

- For the first phase, financial data of 58 Indian stocks and 59 U.S. stocks was collected.
- For the second phase, financial data of 60 global coutry indices was collected.
- Weighted price was considered after forward and backward filling of the data.

$$Weighted\ Price = (High + Low + Open + Close)/4$$

- Daily logarithmic returns were considered for smoothening of the process.

$$Ri(t) = ln\ Pi(t) / Pi(t-1)[]$$

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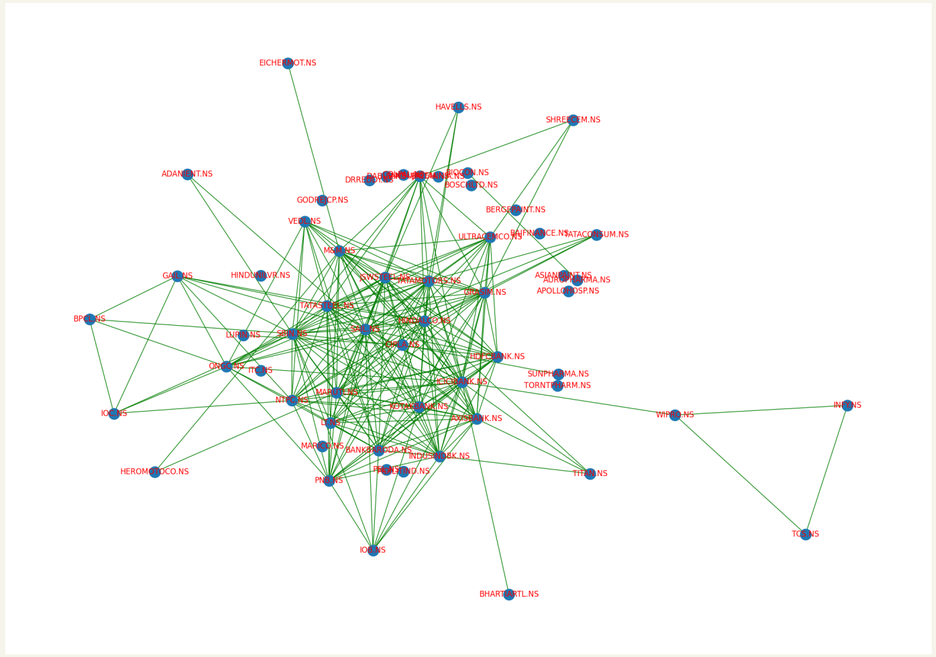
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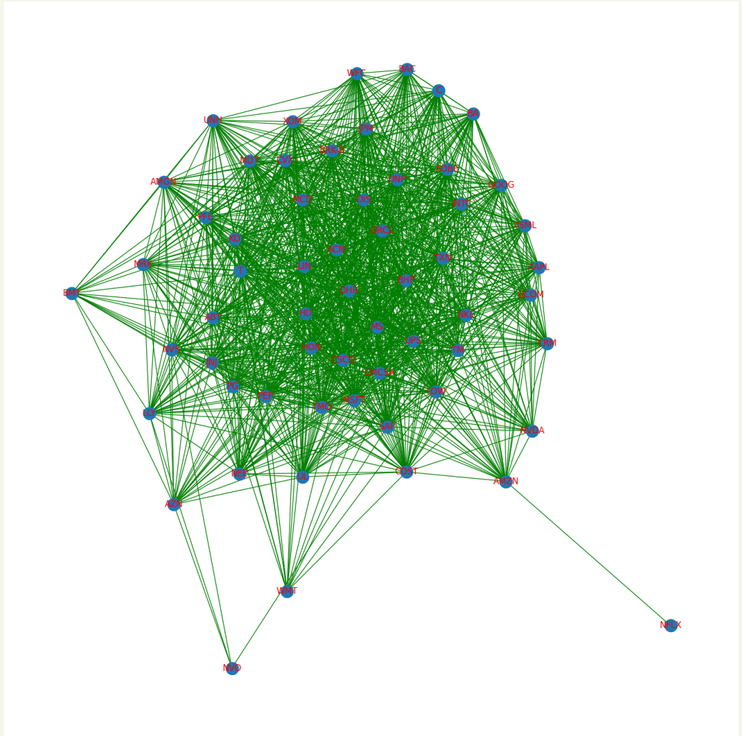
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Network Construction

- Correlation matrix is used for constructing adjacency matric.
- A threshold parameter of 0.36 was considered to consider an edge.



INDIA



USA

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Topological Methods Computation

- 4 Network properties are calculated to see the effects of financial crisis, pandemics, economic and political wars.

Graph Centralities Calculated

- Nodes (stocks) are ranked by the average of percentile ranks of 5 centralities.
- Most influential stocks are found out from this ranking,

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Constructing MST

- Correlation matrix of global indices is found to form MST.
- MST weight is plotted over the period 2018-2025 as weight is a good indicator of level of synchronization in global stock market.
- Following MSTs are plotted for different time periods:
  - Pre COVID-19
  - During COVID-19
  - Post COVID-19
  - Russia-Ukraine War

  to see the regional shifts between countries due to natural, economic or political changes

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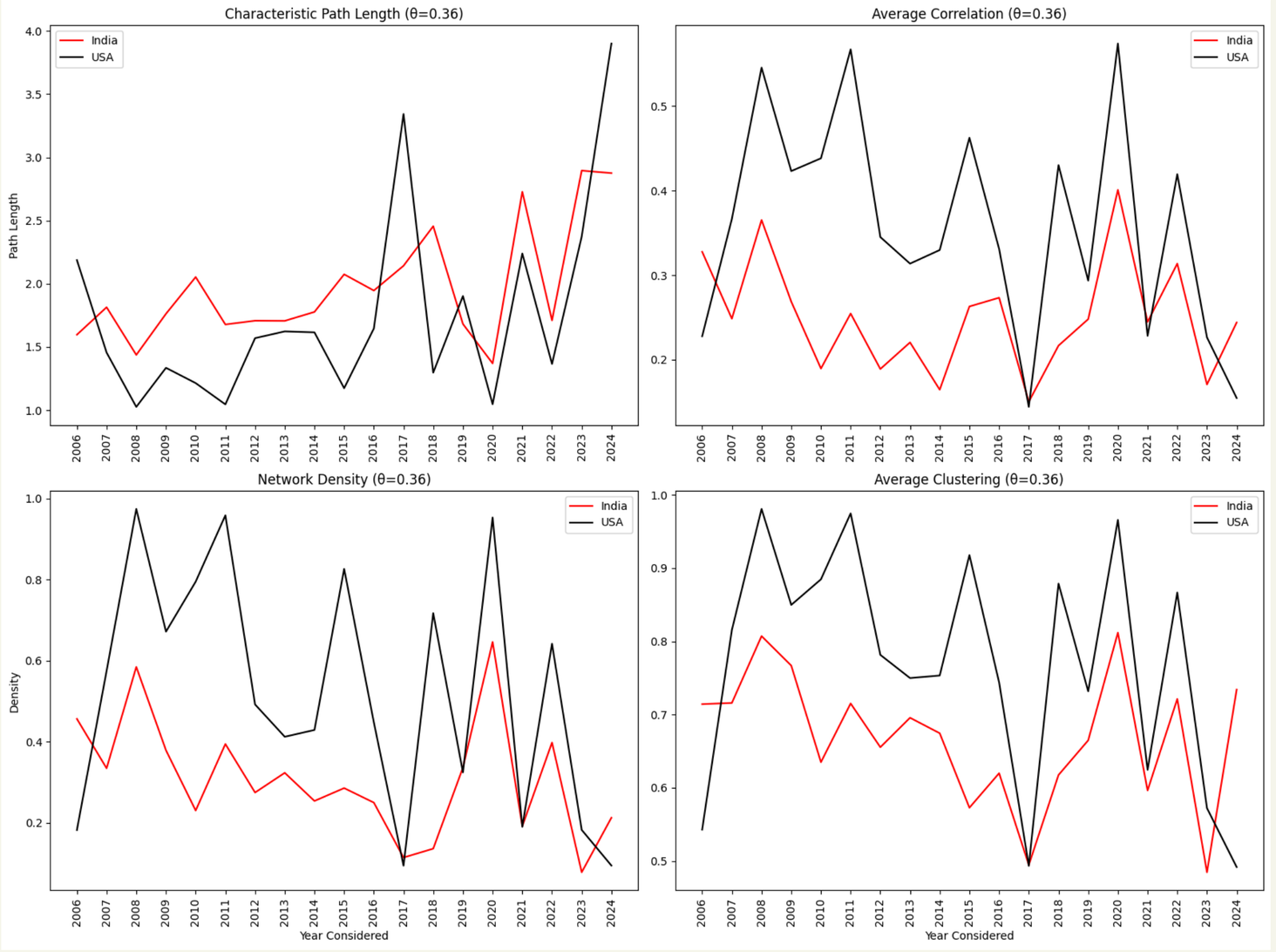
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- Booms in plots of Average Correlation, Network Density, Average Clustering, and a fall in path length can be seen during 2007-08, 2019-20 and 2022.



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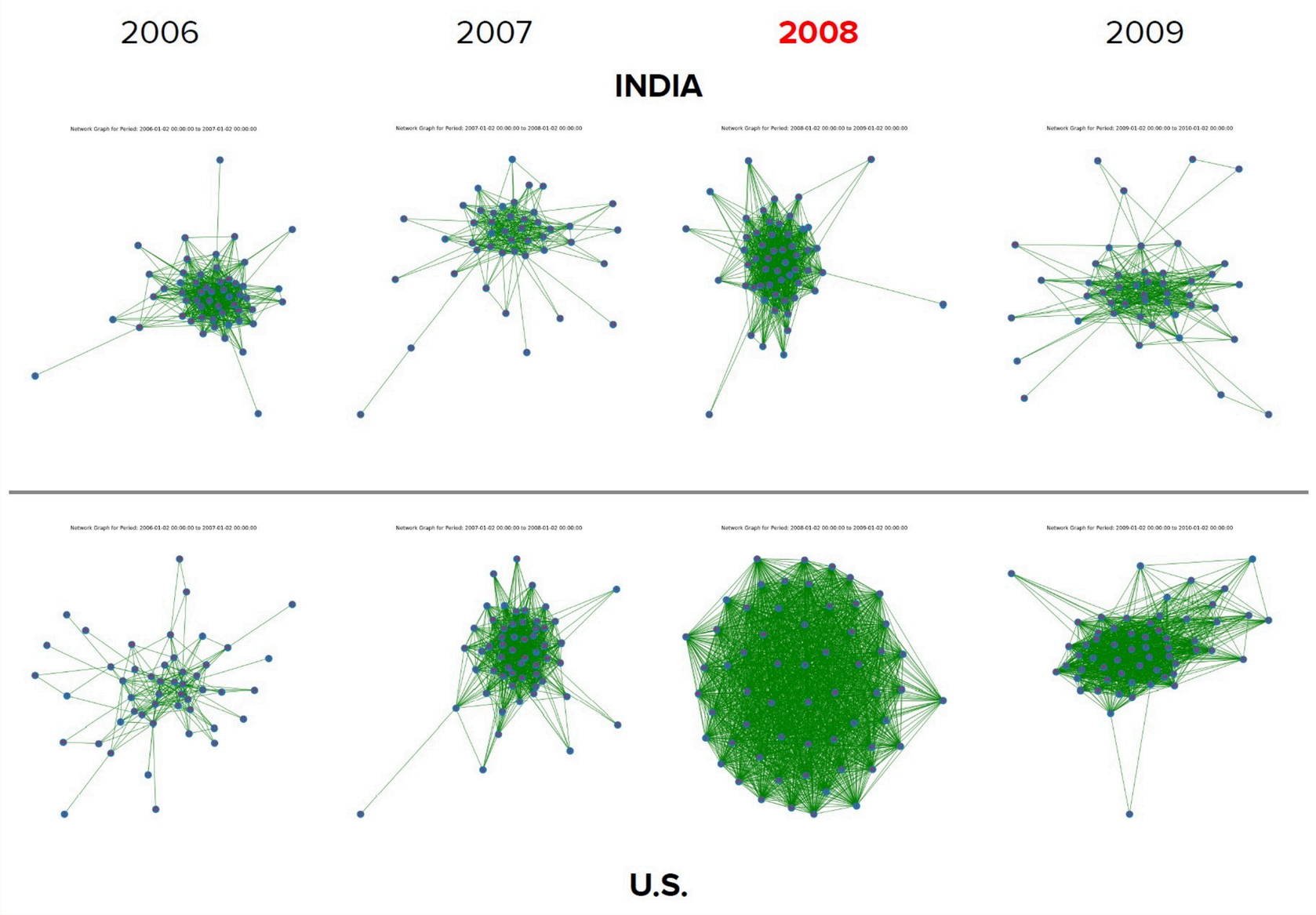
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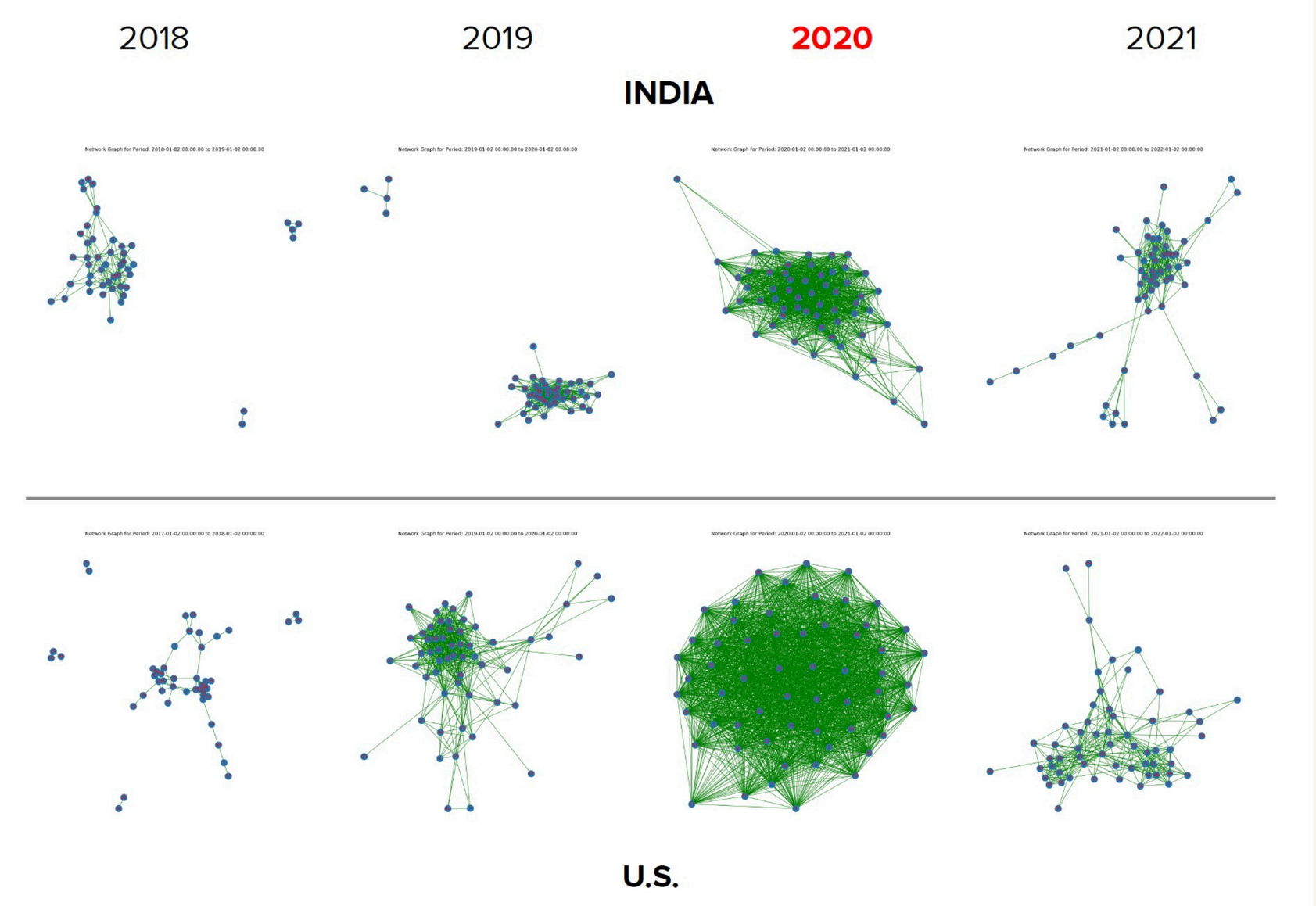
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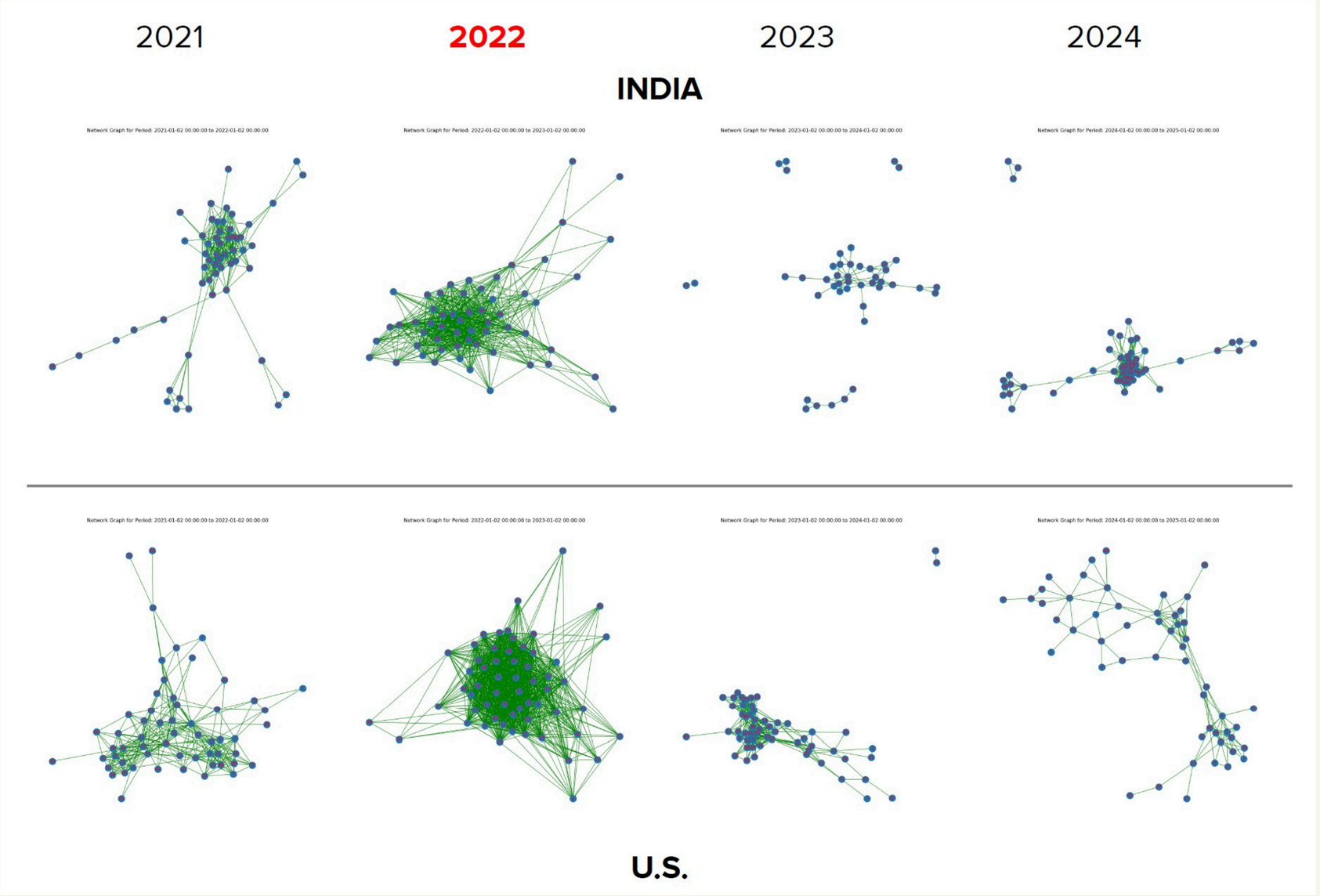
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- The most influential stocks in both regions ranked by the graph centralities.

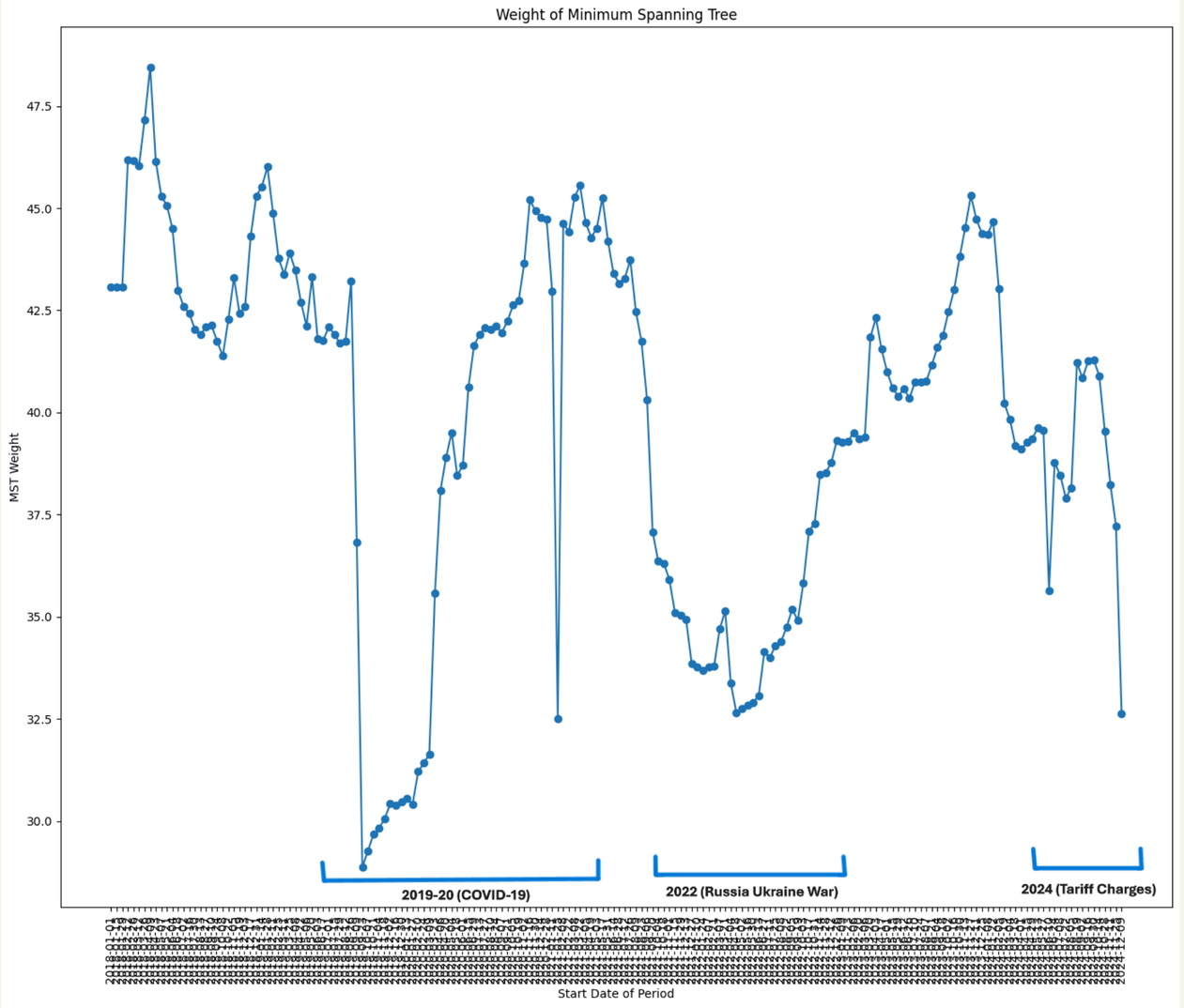
INDIA	USA
State Bank of India	The Home Depot, Inc.
ICICI Bank Limited	Honeywell International Inc.
Hindalco Industries Limited	Cisco Systems, Inc.
Steel Authority of India Limited	Linde plc
Tata Motors Limited	Microsoft Corporation
JSW Steel Limited	Accenture plc
Grasim Industries Limited	Comcast Corporation
Tata Steel Limited	Danaher Corporation
Axis Bank Limited	Thermo Fisher Scientific Inc.

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- Drops in the weights of MST showed sudden drops during Pandemic, beg. of Russian-Ukrainian War and the latter half of 2024.

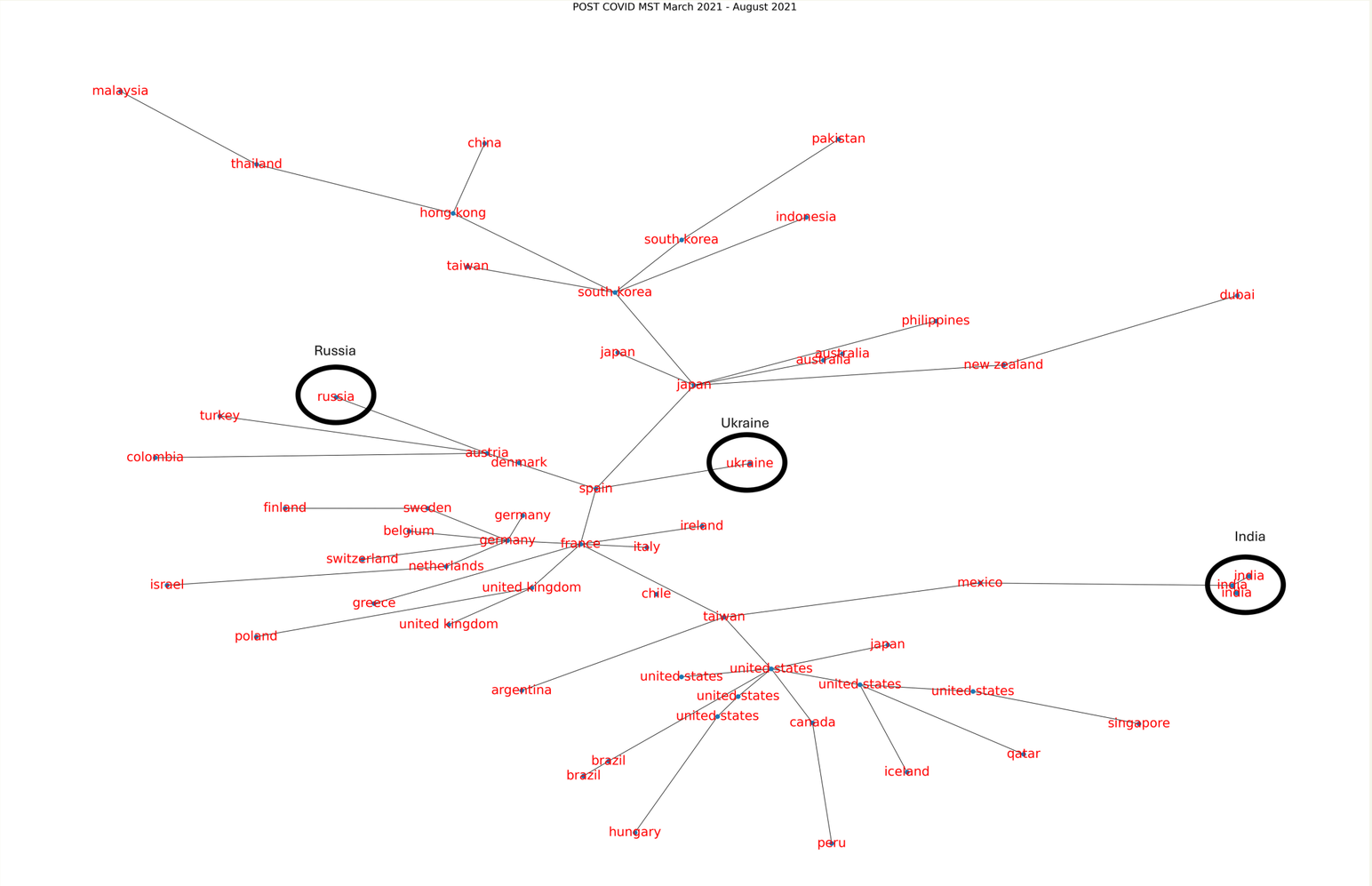


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- Before Pandemic, India had ties with Asian markets but post COVID, India showed ties with American and EU cluster.
- Before 2022, Russia and Ukraine both has ties with EU cluster.



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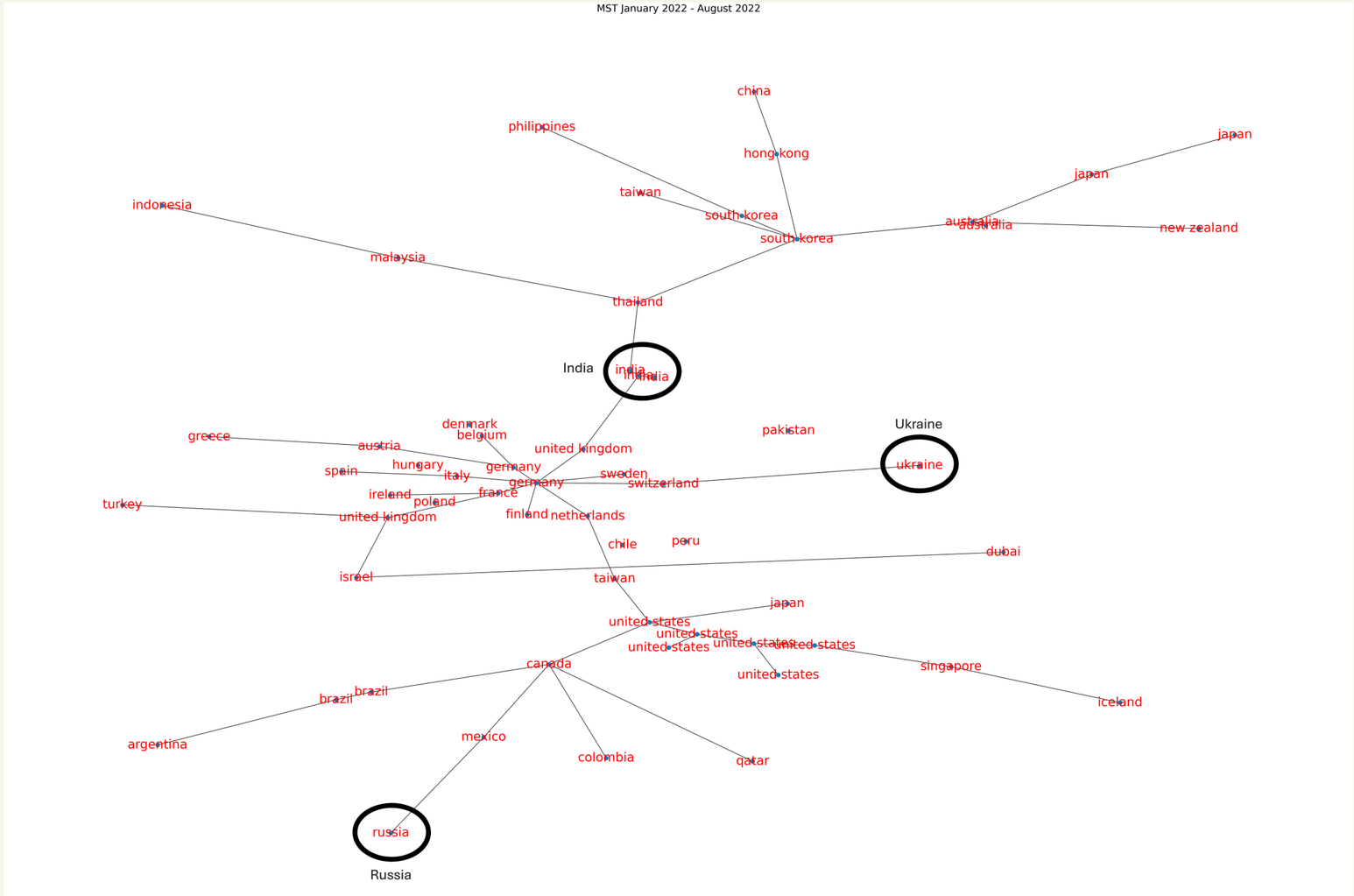
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- India again started to migrate towards Asian markets as the market started recovering post-COVID-19.
- During the war, Russia lost all the trades with EU cluster.



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Together, all these findings tell us that:

- Social network analysis and the theory of threshold networks can be applied to study the interrelationships of global financial markets
- MST weight serves as a real-time cohesion metric for global financial markets.
- Graph centralities and network properties accurately reflect the stability of the financial markets.
- Cluster migrations mirror underlying economic, political changes, and policy shifts.



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THANK YOU